Aobo Li

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Education

Boston University — **Boston, MA, US**
Ph.D., Physics — *September 2015 - September 2020*
"The Tao and Zen of Neutrinos: Neutrinoless Double Beta Decay in KamLAND-Zen 800",
Advisor: Christopher P. Grant

University of Washington, Seattle — **Seattle, WA, US**
B.S., Comprehensive Physics — *September 2011 - September 2015*

Academic Appointments

UC San Diego — **La Jolla, CA, US**
Assistant Professor,
Halıcıoğlu Data Science Institute — *September 2023–present*
Department of Physics

UNC Chapel Hill — **Chapel Hill, NC, US**
COSMS Fellow, — *September 2020–August 2023*
Department of Physics and Astronomy

Boston University — **Boston, MA, US**
Graduate Research Assistant, — *January 2017 - September 2020*
Department of Physics

University of Washington — **Seattle, WA, US**
Undergraduate Research Assistant,
Center for Experimental Nuclear Physics and Astrophysics (CENPA)

Honors and Awards

NSF CAREER Award — **PI [$693,430]**
CAREER: Detecting Extremely Rare Physics Events in the Era of AI for Science

Dissertation Award in Nuclear Physics — **American Physical Society**
`https://www.aps.org/programs/honors/prizes/nuclear.cfm`
'For the invention of a novel machine learning algorithm that broke down significant technological barriers with monolithic liquid scintillator detectors and, in turn, delivered the world's most sensitive search for neutrinoless double beta decay."

Postdoctoral Award of Research Excellence — **UNC Chapel Hill**
`https://research.unc.edu/2022/09/14/2022-pare-award-recipients/`

MLST Paper Award — **NeurIPS 2022 ML4PS Workshop**
`https://ml4physicalsciences.github.io/2022/`

Competitive Research Funding

DOE Genesis Mission Phase I Award — **Lead PI [$719,000]**
Denoising Quantum Sensors: Agentic Algorithm Creation and Real-Time AI for Quantum-Enabled Rare Event Discovery

DOE Genesis Mission Phase I Award — **Co-PI [$104,021]**
Enabling Rare Event Discovery with Surrogate Modeling and Simulation AI Agents

DOE Genesis Mission Phase I Award — **Co-PI [$164,000]**
Developing a Foundation Model to Unify Neutrinoless Double Beta Decay Search Technologies

DOE Genesis Mission Phase I Award — **Co-PI [$106,622]**
Low-latency Embedded AI for Physics - From Waveforms to discovery

DOE AI and ML Applied to Nuclear Science and Technology — **Co-PI [$320,000]**

New Approaches to Bayesian Uncertainty Quantification for Nuclear Science

Professional Activities

Department of Energy **2024-Present**
Invited Participant of HEP&NP AI Roundtable

Department of Energy, National Science Foundation **2022-Present**
reviewer

National Nuclear Physics Summer School **2024,2026**
Invited Lecturer on AI/Machine Learning

PRL,PRX, PRC, PRD, EPJC, CPC, NST, MLST **2023-Present**
Referee

NeurIPS, KDD **2025,2026**
Reviewer

Accelerated AI Algorithms for Data-Driven Discovery (A3D3) **2023-Present**
Education & Outreach Undergraduate Activity Chair

Conference Organization

Fast Machine Learning for Science **2026**
Technical Program Committee Chair

APS Global Physics Summit **2026**
Program Committee Co-Chair, Group of Data Science

$0\nu\beta\beta$ AI Summer School **2026**
Organizing and Steering Committee Chair

Neutrino Physics and Machine Learning Conference **2023,2024,2025,2026**
Organizing Committee Member

APS DNP **2023, 2025**
Organizer of AI in Nuclear Physics Experiment Workshop

Summer Bootcamp of Deep Learning and Its Applicaitons **2024**
Organizer and Lecturer

A3D3 Undergraduate Research Symposium **2024**
Organizer

INVITED ACADEMIC TALKS

- SIDERIUS: Resource-Aware AI Agents for Particle and Nuclear Physics, Invited Talk, Conference on the Application of Accelerators in Research and Industry (CAARI), August 2026
- Artificial Intelligence for the Search of Rare Astrophysical Events, Invited Plenary Talk, APS Global Physics Summit, March 2026
- The 3 Pillars for the Search of Rare Astrophysical Events, Invited Talk, 10th OCPA International Conference, June 2026
- Searching for Rare Events with Rare AI, Invited GPPU Lecture, Tohoku University, March 2025
- Algorithm, Data, and Hardware: The Three AI Thrusts That Accelerate Neutrinoless Double-Beta Decay Searches, Invited Talk, APS Division of Nuclear Physics Fall Meeting, October 2024
- Machine Learning for the Detection of Rare Events, Invited Plenary Talk, 22nd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT 2024), March 2024
- Pushing Rare Event Searches to the Limit with Machine Learning Algorithms, Invited Plenary Talk, Conference on Science at the Sanford Underground Research Facility (CoSSURF 2024), May 2024

- LEGEND: The Road Onward to the Inverted Mass Ordering Region, Invited Plenary Talk, Symposium on Frontiers of Underground Physics, November 2023
- Building Efficient and Interpretable AI for Neutrinoless Double-Beta Decay Searches, Invited Talk, APS Division of Nuclear Physics Award Session, October 2022

Selected Publications

J. T. Fry, X. H. Fu, Z. Fu, K. M. W. Pappas, L. Winslow, and **A. Li**, "TIDMAD: Time Series Dataset for Discovering Dark Matter with AI Denoising," *Proceedings of the Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025. **Spotlight Paper**.

A.-K. Schuetz, A. W. P. Poon, and **A. Li**, "RESuM: A Rare Event Surrogate Model for Physics Detector Design," *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR)*, 2025. **Spotlight Paper**.

A. Li, Z. Fu, L. A. Winslow, C. Grant, H. Song, H. Ozaki, I. Shimizu, and A. Takeuchi, "KamNet: An Integrated Spatiotemporal Deep Neural Network for Rare Event Searches in KamLAND-Zen," *Physical Review C* **107**, 014323 (2023). **Editors' Suggestion**.

I. J. Arnquist *et al.*, "Interpretable Boosted-Decision-Tree Analysis for the Majorana Demonstrator," *Physical Review C* **107**, 014321 (2023).

K. Bhimani, J. Gruszko, M. Clark, J. Wilkerson, and **A. Li**, "CycleGAN-Driven Transfer Learning for Electronics Response Emulation in High-Purity Germanium Detectors," *Machine Learning: Science and Technology* **7**, 015009 (2026).

Z. Fu, C. Grant, D. M. Krawiec, **A. Li**, and L. A. Winslow, "Generative Models for Simulation of KamLAND-Zen," *European Physical Journal C* **84**, 651 (2024).

E. León, **A. Li**, M. A. Bahena Schott, *et al.*, "Machine Learning-Powered Data Cleaning for LEGEND: A Semi-Supervised Approach Using Affinity Propagation and Support Vector Machines," *Machine Learning: Science and Technology* **6**, 015064 (2025).

I. J. Arnquist *et al.*, "Majorana Demonstrator Data Release for AI/ML Applications," arXiv:2308.10856 (2023).

S. Abe *et al.* (KamLAND-Zen Collaboration), "Search for the Majorana Nature of Neutrinos in the Inverted Mass Ordering Region with KamLAND-Zen," *Physical Review Letters* **130**, 051801 (2023). **Featured in Physics**; **Editors' Suggestion**.

S. Abe *et al.* (KamLAND-Zen Collaboration), "Search for Majorana Neutrinos with the Complete KamLAND-Zen Dataset," *Physical Review Letters* **135**, 262501 (2025).

I. J. Arnquist *et al.* (Majorana Collaboration), "Final Result of the Majorana Demonstrator's Search for Neutrinoless Double-β Decay in ^{76}Ge," *Physical Review Letters* **130**, 062501 (2023). **Editors' Suggestion**.

H. Acharya *et al.* (LEGEND Collaboration), "First Results on the Search for Lepton Number Violating Neutrinoless Double-β Decay with the LEGEND-200 Experiment," *Physical Review Letters* **136**, 022701 (2026).

Complete Publication List

[1] Kevin Bhimani, Julieta Gruszko, Morgan Clark, John Wilkerson, and Aobo Li. Cyclegan-driven transfer learning for electronics response emulation in high-purity germanium detectors. *Machine Learning: Science and Technology*, 7(1):015009, jan 2026.

[2] H. Acharya et al. First results on the search for lepton number violating neutrinoless double-β decay with the legend-200 experiment. *Phys. Rev. Lett.*, 136:022701, Jan 2026.

[3] J. T. Fry, Xinyi Hope Fu, Zhenghao Fu, Kaliroë Mabelle West Pappas, Lindley Winslow, and Aobo Li. TIDMAD: Time series dataset for discovering dark matter with AI denoising. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2025. **Spotlight Paper**.

[4] Ann-Kathrin Schuetz, A.W.P. Poon, and Aobo Li. RESum: A rare event surrogate model for physics detector design. In *The Thirteenth International Conference on Learning Representations*, 2025. **Spotlight Paper**.

[5] S. Abe et al. Search for majorana neutrinos with the complete kamland-zen dataset. *Phys. Rev. Lett.*, 135:262501, Dec 2025.

[6] E León, A Li, M A Bahena Schott, B Bos, M Busch, J R Chapman, G L Duran, J Gruszko, R Henning, E L Martin, and J F Wilkerson. Machine learning-powered data cleaning for legend: a semi-supervised approach using affinity propagation and support vector machines. *Machine Learning: Science and Technology*, 6(1):015064, mar 2025.

[7] J. Aalbers et al. Neutrinoless double beta decay sensitivity of the xlzd rare event observatory. *J. Phys. G*, 52(4):045102, 2025, 2410.19016.

[8] J. Aalbers et al. The xlzd design book: towards the next-generation liquid xenon observatory for dark matter and neutrino physics. *Eur. Phys. J. C*, 85(10):1192, 2025, 2410.17137.

[9] Zhenghao Fu, Christopher Grant, Dominika M. Krawiec, Aobo Li, and Lindley A. Winslow. Generative models for simulation of kamland-zen. *Eur. Phys. J. C*, 84(6):651, 2024, 2312.14372.

[10] I. J. Arnquist et al. Search for charge non-conservation and pauli exclusion principle violation with the majorana demonstrator. *Nature Physics*, pages 1–6, 2024.

[11] A. Li, Z. Fu, L. A. Winslow, C. Grant, H. Song, H. Ozaki, I. Shimizu, and A. Takeuchi. Kamnet: An integrated spatiotemporal deep neural network for rare event searches in kamland-zen*. *Phys. Rev. C*, 107(1):014323, 2023, 2203.01870. **Editor's Suggestion**.

[12] S. Abe et al. Search for the majorana nature of neutrinos in the inverted mass ordering region with kamland-zen. *Phys. Rev. Lett.*, 130(5):051801, 2023, 2203.02139. **Featured in Physics**, **Editor's Suggestion**.

[13] I. J. Arnquist et al. Final result of the majorana demonstrator's search for neutrinoless double-β decay in ^{76}Ge. *Phys. Rev. Lett.*, 130:062501, Feb 2023. **Editor's Suggestion**.

[14] I. J. Arnquist, F. T. Avignone, A. S. Barabash, C. J. Barton, K. H. Bhimani, et al. Interpretable boosted-decision-tree analysis for the majorana demonstrator. *Phys. Rev. C*, 107:014321, Jan 2023.

[15] I. J. Arnquist et al. Majorana demonstrator data release for ai/ml applications. 8 2023, 2308.10856.

[16] S. Abe et al. Measurement of cosmic-ray muon spallation products in a xenon-loaded liquid scintillator with kamland. *Phys. Rev. C*, 107(5):054612, 2023, 2301.09307.

[17] S. Abe et al. First measurement of the strange axial coupling constant using neutral-current quasielastic interactions of atmospheric neutrinos at kamland. *Phys. Rev. D*, 107(7):072006, 2023, 2211.13911.

[18] I. J. Arnquist et al. Energy calibration of germanium detectors for the majorana demonstrator. 6 2023, 2306.08661.

[19] S. Abe et al. Search for supernova neutrinos and constraint on the galactic star formation rate with the kamland data. 4 2022, 2204.12065.

[20] S. Abe et al. Abundances of uranium and thorium elements in earth estimated by geoneutrino spectroscopy. *Geophysical Research Letters*, 49(16):e2022GL099566, 2022.

[21] N. Abgrall et al. The majorana demonstrator readout electronics system. *JINST*, 17(05):T05003, 2022, 2111.09351.

[22] I. J. Arnquist et al. Exotic dark matter search with the majorana demonstrator. 6 2022, 2206.10638.

[23] I. J. Arnquist et al. Search for solar axions via axion-photon coupling with the majorana demonstrator. *Phys. Rev. Lett.*, 129(8):081803, 2022, 2206.05789.

[24] A. Li, J. Gruszko, B. Bos, T. Caldwell, E. León, and J. Wilkerson. Ad-hoc pulse shape simulation using cyclic positional u-net. *Proceedings of NeurIPS 2022 Machine Learning for Physical Science Workshop*, Dec 2022. `https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_12.pdf`.

[25] I. J. Arnquist et al. Charge trapping and energy performance of the majorana demonstrator. 8 2022, 2208.03424.

[26] S. Abe et al. Search for solar flare neutrinos with the kamland detector. *Astrophys. J.*, 924(2):103, 2022, 2105.02458.

[27] S. Abe et al. Search for correlated low-energy electron antineutrinos in kamland with gamma-ray bursts. *Astrophys. J.*, 927(1):69, 2022, 2112.04918.

[28] Joshua R. Klein et al. Future advances in photon-based neutrino detectors: A snowmass white paper. 3 2022, 2203.07479.

[29] I. J. Arnquist et al. Search for pauli exclusion principle violation by pair production in the majorana demonstrator calibration data. 3 2022, 2203.02033.

[30] I. J. Arnquist et al. The study of ^{13}c(α,n)^{16}o reactions in the majorana demonstrator calibration data. *Phys. Rev. C*, 105(1):064610, 2022, 2203.14228.

[31] I. J. Arnquist et al. Search for spontaneous radiation from wavefunction collapse in the majorana demonstrator. *Phys. Rev. Lett.*, 129(8):080401, 2022, 2202.01343.

[32] N. Abgrall et al. The large enriched germanium experiment for neutrinoless $\beta\beta$ decay: Legend-1000 preconceptual design report. 7 2021, 2107.11462.

[33] V. Albanese et al. The sno+ experiment. *JINST*, 16(08):P08059, 2021, 2104.11687.

[34] Y. Gando et al. The nylon balloon for xenon loaded liquid scintillator in kamland-zen 800 neutrinoless double-beta decay search experiment. *JINST*, 16(08):P08023, 2021, 2104.10452.

[35] M. R. Anderson et al. Development, characterisation, and deployment of the sno+ liquid scintillator. *JINST*, 16:P05009, 2021, 2011.12924.

[36] M. R. Anderson et al. Optical calibration of the sno+ detector in the water phase with deployed sources. *JINST*, 16(10):P10021, 2021, 2106.03951.

[37] Aobo Li. Kamland-zen 800 status and prospect with the artificial intelligence powered analysis. *J. Phys. Conf. Ser.*, 2156(1):012114, 2021.

[38] S. Abe et al. Search for low-energy electron antineutrinos in kamland associated with gravitational wave events. *Astrophys. J.*, 909(2):116, 2021, 2012.12053.

[39] S. Abe et al. A search for charged excitation of dark matter with the kamland-zen detector. 1 2021, 2101.06049.

[40] Aobo Li. A bayesian approach to neutrinoless double beta decay analysis in KamLAND-zen. *Journal of Physics: Conference Series*, 1468:012201, feb 2020.

[41] M.R. Anderson et al. Measurement of neutron-proton capture in the sno+ water phase. *Phys. Rev. C*, 102(1):014002, 2020, 2002.10351.

[42] A. Li, A. Elagin, S. Fraker, C. Grant, and L. Winslow. Suppression of cosmic muon spallation backgrounds in liquid scintillator detectors using convolutional neural networks. *Nucl. Instrum. Meth. A*, 947:162604, 2019, 1812.02906.

[43] M. Anderson et al. Measurement of the ^{8}B solar neutrino flux in SNO+ with very low backgrounds. *Phys. Rev. D*, 99:012012, Jan 2019.

[44] M. Anderson et al. Search for invisible modes of nucleon decay in water with the sno+ detector. *Phys. Rev.*, D99(3):032008, 2019, 1812.05552.

[45] N. Abgrall et al. High voltage testing for the majorana demonstrator. *Nucl. Instrum. Meth.*, A823:83–90, 2016, 1603.08483.